

MUHAMMAD HUSSAIN KHAN LODHI

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Summary

Software Engineer and full-stack MERN developer with a strong foundation in data structures, object-oriented design, and database systems. Experienced in designing RESTful APIs, JWT authentication, role-based access control, database-backed applications, deterministic recommendation engines, and automated testing with React, Node.js, Express, MongoDB, and MySQL. IBM-certified in full-stack JavaScript development; BSCS graduate with a 3.7/4.0 CGPA.

Experience

NextSkill Institute

Full Stack Development Trainee

Jun 2026 - Sep 2026

Karachi, Pakistan

- Completed a NAVTTC-affiliated Full Stack Development training program with an IBM-aligned curriculum, progressing from HTML5/CSS3/JavaScript fundamentals to React-based multi-page and e-commerce applications
- Built and version-controlled a series of front-end projects, including a personal portfolio, an expense tracker, and a multi-page e-commerce site, using Git/GitHub for iterative submission and review

10Pearls - 10Shine Program

MERN Stack Development Intern

Certificate | Apr 2026 - May 2026

Remote, Karachi

- Designed and implemented the data model and REST API for a full-stack Notes application (React, Node.js, Express, MySQL), including JWT authentication, email verification, and full CRUD (add, update, delete, search)
- Authored unit/integration tests (Mocha, Chai, NYC coverage) and resolved issues flagged by SonarQube static analysis; collaborated via Git/GitHub with pull requests and code review

Self-Directed Learning & Development

Independent Study

Sep 2022 - Present

Karachi, Pakistan

- Designed, built, and deployed 10+ full-stack MERN applications (React, Node.js, MongoDB/MySQL) independently, applying data structures, OOP, and database design coursework to real projects
- Practiced collaborative software workflows (Git/GitHub) and iterative, test-driven development

Projects

Smart Complaint Management System (MERN Stack)

GitHub | Live Demo

- Designed a role-based access-control system and a two-stage state machine, covering account approval and complaint lifecycle (Pending, In Progress, Resolved/Rejected), enforced across 15 REST endpoints
- Built the data model and API layer (React, Node.js, Express, MongoDB/Mongoose) with JWT auth, bcrypt hashing, and admin-only route guards; deployed frontend and backend as independent Vercel projects

Notes Management Platform (Full-Stack)

GitHub | Live Demo

- Architected a full-stack notes application (React, Node.js, Express, MySQL) with user registration, login, JWT-based authentication, and email verification workflows
- Engineered full CRUD operations with title/content search, automated endpoint testing, and independent Vercel deployments with cloud MySQL (TiDB Cloud) integration

SHOP.CO E-Commerce Platform (Full-Stack)

GitHub | Live Demo

- Engineered a pixel-faithful MERN e-commerce platform with robust REST endpoints supporting multi-filter product search, pagination, dynamic category browsing, and secure order processing
- Integrated MongoDB/Mongoose data models for products, cart items, user reviews, and promo codes, coupled with a responsive React and Vite frontend deployed on Vercel

Education

Sindh Madressatul Islam University

Bachelor of Science in Computer Science; CGPA: 3.7/4.0

Sep 2022 - Jun 2026

Karachi, Pakistan

- Relevant Coursework:** Data Structures, Object-Oriented Programming, Database Systems, Software Engineering, Computer Networks, Operating Systems, Mobile App Development

Certifications

IBM Full-Stack JavaScript Developer (Professional Certificate)

Verify | Sep 2026

Technical Skills

Languages: JavaScript, TypeScript, Python, Java, C++, C, SQL

Core Computer Science: Data Structures, Object-Oriented Programming, REST API Design, Database Design, Unit Testing

Frontend Technologies: HTML5, CSS3, JSX, React.js, Next.js, React Router, Tailwind CSS, Context API, Redux Toolkit, Vite

Backend Technologies: Node.js, Express.js, REST APIs, Socket.IO, JWT Authentication, CORS, File Uploads

Databases and ORMs: MongoDB, MongoDB Atlas, Mongoose, PostgreSQL, Sequelize, MySQL, Turso (libSQL), SQL Server

Developer Tools: Git, GitHub, GitHub Actions, Docker, ESLint, Postman, SonarQube, Mocha/Chai, VS Code, NetBeans

Cloud and Deployment: Vercel, GitHub Pages, Railway, Netlify

Soft Skills: Problem-Solving, Teamwork, Communication, Adaptability, Time Management